

C_Leddq i1_i2_i3_i4_ →

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\overline{y}_d^{i4i3} y_e^{i1i2} \overline{a}_e^{pr} a_e^{pr} \text{LF}_{3,1,0} [m_l^p, m_e^r] - 3 \overline{y}_d^{i4i3} y_e^{i1i2} \overline{a}_e^{pr} a_e^{pr} \text{LF}_{4,1,-1} [m_l^p, m_e^r] + \right. \\
& 2 \overline{y}_d^{i4i3} y_e^{i1i2} \overline{a}_e^{pr} a_e^{pr} \text{LF}_{5,1,-2} [m_l^p, m_e^r] + 3 \overline{y}_d^{i4i3} y_e^{i1i2} \overline{a}_d^{pr} a_d^{pr} \text{LF}_{3,1,0} [m_q^p, m_d^r] - \\
& 9 \overline{y}_d^{i4i3} y_e^{i1i2} \overline{a}_d^{pr} a_d^{pr} \text{LF}_{4,1,-1} [m_q^p, m_d^r] + 6 \overline{y}_d^{i4i3} y_e^{i1i2} \overline{a}_d^{pr} a_d^{pr} \text{LF}_{5,1,-2} [m_q^p, m_d^r] + \\
& 3 \tilde{\mu}^2 \overline{y}_d^{i4i3} y_e^{i1i2} \overline{y}_u^{pr} y_u^{pr} \text{LF}_{3,1,0} [m_u^r, m_q^p] - 9 \tilde{\mu}^2 \overline{y}_d^{i4i3} y_e^{i1i2} \overline{y}_u^{pr} y_u^{pr} \text{LF}_{4,1,-1} [m_u^r, m_q^p] + \\
& 6 \tilde{\mu}^2 \overline{y}_d^{i4i3} y_e^{i1i2} \overline{y}_u^{pr} y_u^{pr} \text{LF}_{5,1,-2} [m_u^r, m_q^p] - 2 g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{3,1,-1} [\tilde{\mu}, m_1] + \\
& 4 g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] - 2 g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] - \\
& 6 g_2^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{3,1,-1} [\tilde{\mu}, m_2] + 12 g_2^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] - \\
& 6 g_2^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] + \frac{1}{36} m_1 g_1^2 y_e^{i1i2} \overline{a}_d^{i4i3} \text{LF}_{2,2,1,-1} [m_1, m_d^{i3}, m_q^{i4}] - \frac{1}{4} m_1 g_1^2 \\
& \overline{y}_d^{i4i3} a_e^{i1i2} \text{LF}_{2,2,1,-1} [m_1, m_e^{i2}, m_l^{i1}] - \frac{1}{4} m_1 g_1^2 \overline{y}_d^{i4i3} a_e^{i1i2} \text{LF}_{2,2,1,-1} [m_1, m_l^{i1}, m_e^{i2}] + \\
& \frac{1}{36} m_1 g_1^2 y_e^{i1i2} \overline{a}_d^{i4i3} \text{LF}_{2,2,1,-1} [m_1, m_q^{i4}, m_d^{i3}] + \frac{1}{6} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_d^{i3}] + \\
& \frac{1}{2} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_e^{i2}] - \frac{1}{4} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_l^{i1}] + \\
& \frac{1}{12} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_q^{i4}] + \frac{3}{4} g_2^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_2, \tilde{\mu}, m_l^{i1}] + \\
& \frac{3}{4} g_2^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_2, \tilde{\mu}, m_q^{i4}] - \frac{2}{3} m_3 g_3^2 y_e^{i1i2} \overline{a}_d^{i4i3} \text{LF}_{2,2,1,-1} [m_3, m_d^{i3}, m_q^{i4}] - \\
& \frac{2}{3} m_3 g_3^2 y_e^{i1i2} \overline{a}_d^{i4i3} \text{LF}_{2,2,1,-1} [m_3, m_q^{i4}, m_d^{i3}] - \frac{1}{18} m_1 g_1^2 y_e^{i1i2} \overline{a}_d^{i4i3} \\
& \text{LF}_{2,1,1,0} [m_d^{i3}, m_1, m_q^{i4}] + \frac{1}{18} m_1 g_1^2 y_e^{i1i2} \overline{a}_d^{i4i3} \text{LF}_{3,1,1,-1} [m_d^{i3}, m_1, m_q^{i4}] + \\
& \frac{4}{3} m_3 g_3^2 y_e^{i1i2} \overline{a}_d^{i4i3} \text{LF}_{2,1,1,0} [m_d^{i3}, m_3, m_q^{i4}] - \frac{4}{3} m_3 g_3^2 y_e^{i1i2} \overline{a}_d^{i4i3} \\
& \text{LF}_{3,1,1,-1} [m_d^{i3}, m_3, m_q^{i4}] + \frac{1}{2} m_1 g_1^2 \overline{y}_d^{i4i3} a_e^{i1i2} \text{LF}_{2,1,1,0} [m_e^{i2}, m_1, m_l^{i1}] - \\
& \frac{1}{2} m_1 g_1^2 \overline{y}_d^{i4i3} a_e^{i1i2} \text{LF}_{3,1,1,-1} [m_e^{i2}, m_1, m_l^{i1}] + \frac{1}{2} m_1 g_1^2 \overline{y}_d^{i4i3} a_e^{i1i2} \\
& \text{LF}_{2,1,1,0} [m_l^{i1}, m_1, m_e^{i2}] - \frac{1}{2} m_1 g_1^2 \overline{y}_d^{i4i3} a_e^{i1i2} \text{LF}_{3,1,1,-1} [m_l^{i1}, m_1, m_e^{i2}] + \\
& \frac{1}{2} \tilde{\mu}^2 \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} y_u^{pr} \text{LF}_{2,1,1,0} [m_q^p, m_u^r, \tilde{\mu}] - \frac{1}{2} \tilde{\mu}^2 \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} y_u^{pr} \\
& \text{LF}_{3,1,1,-1} [m_q^p, m_u^r, \tilde{\mu}] - \frac{1}{4} \tilde{\mu}^2 \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} y_u^{pr} \text{LF}_{2,2,1,-1} [m_q^p, \tilde{\mu}, m_u^r] - \\
& \frac{1}{18} m_1 g_1^2 y_e^{i1i2} \overline{a}_d^{i4i3} \text{LF}_{2,1,1,0} [m_q^{i4}, m_1, m_d^{i3}] + \frac{1}{18} m_1 g_1^2 y_e^{i1i2} \overline{a}_d^{i4i3} \\
& \text{LF}_{3,1,1,-1} [m_q^{i4}, m_1, m_d^{i3}] + \frac{4}{3} m_3 g_3^2 y_e^{i1i2} \overline{a}_d^{i4i3} \text{LF}_{2,1,1,0} [m_q^{i4}, m_3, m_d^{i3}] - \\
& \frac{4}{3} m_3 g_3^2 y_e^{i1i2} \overline{a}_d^{i4i3} \text{LF}_{3,1,1,-1} [m_q^{i4}, m_3, m_d^{i3}] + \frac{1}{2} \tilde{\mu}^2 \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} y_u^{pr} \\
& \text{LF}_{2,1,1,0} [m_u^r, m_q^p, \tilde{\mu}] - \frac{1}{2} \tilde{\mu}^2 \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} y_u^{pr} \text{LF}_{3,1,1,-1} [m_u^r, m_q^p, \tilde{\mu}] - \\
& \frac{1}{4} \tilde{\mu}^2 \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} y_u^{pr} \text{LF}_{2,2,1,-1} [m_u^r, \tilde{\mu}, m_q^p] - \frac{1}{3} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} \\
& \text{LF}_{1,1,1,1,-1} [m_1, m_d^{i3}, m_e^{i2}, \tilde{\mu}] + \frac{1}{6} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,1,1,-1} [m_1, m_d^{i3}, m_l^{i1}, \tilde{\mu}] - \\
& \frac{1}{6} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,1,1,-1} [m_1, m_e^{i2}, m_q^{i4}, \tilde{\mu}] + \frac{1}{12} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} \\
& \left. \text{LF}_{1,1,1,1,-1} [m_1, m_l^{i1}, m_q^{i4}, \tilde{\mu}] - \frac{3}{4} g_2^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,1,1,-1} [m_2, m_l^{i1}, m_q^{i4}, \tilde{\mu}] \right)
\end{aligned}$$